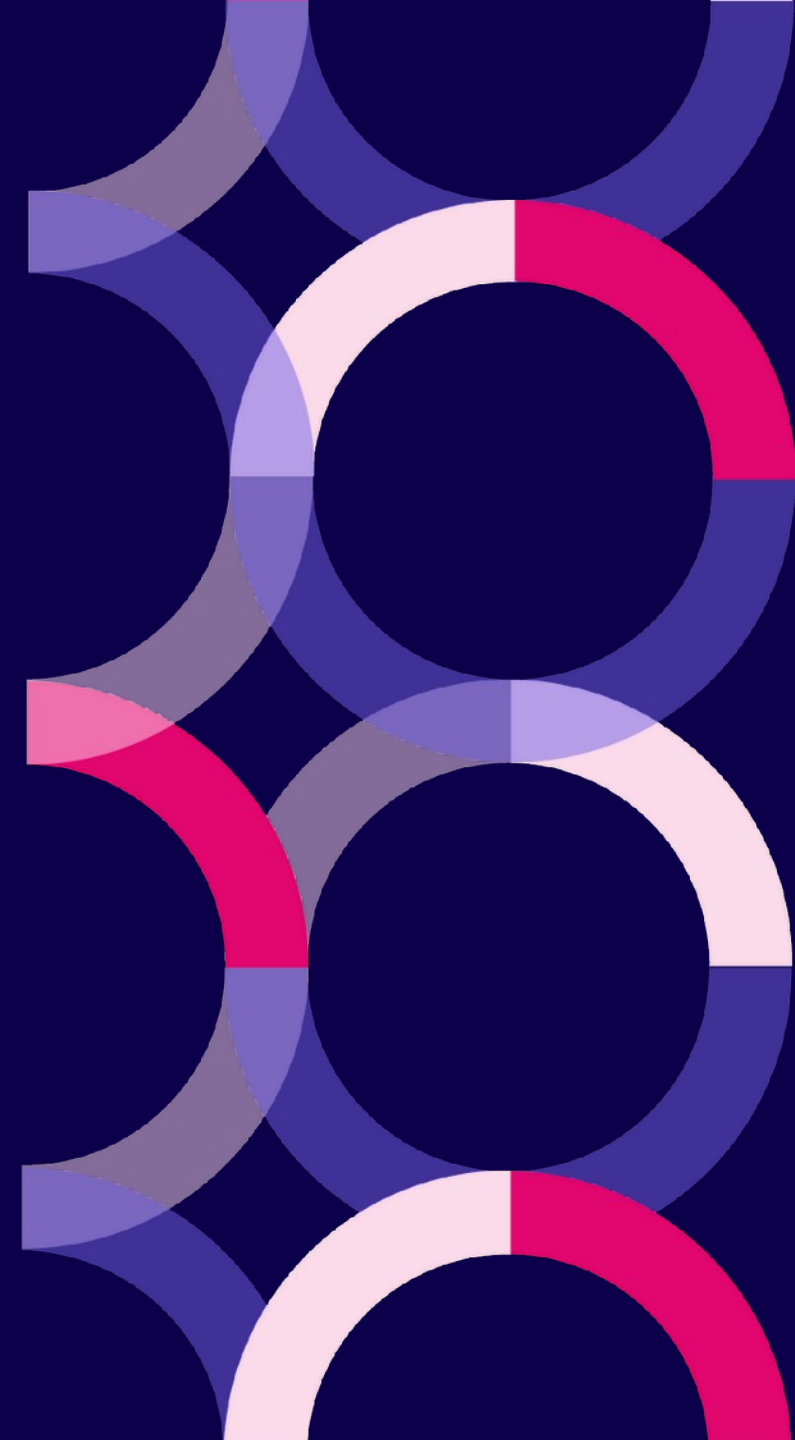


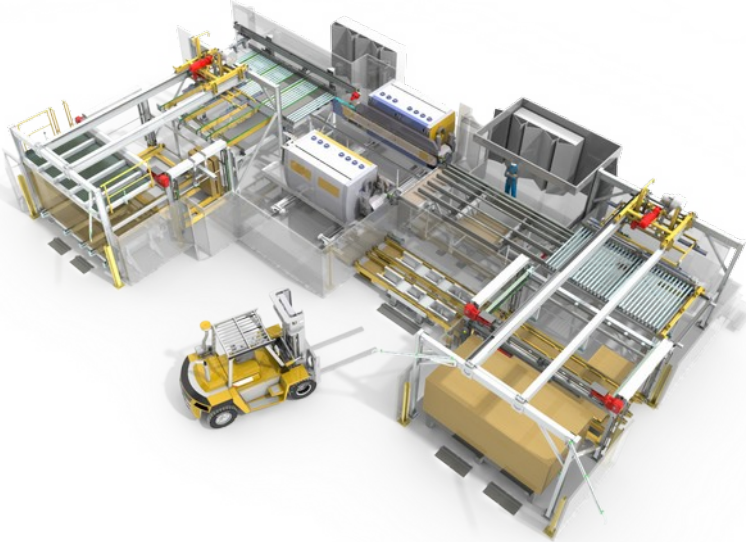
DCS in general

If a PLC can control a machine, what is needed to control an entire factory?



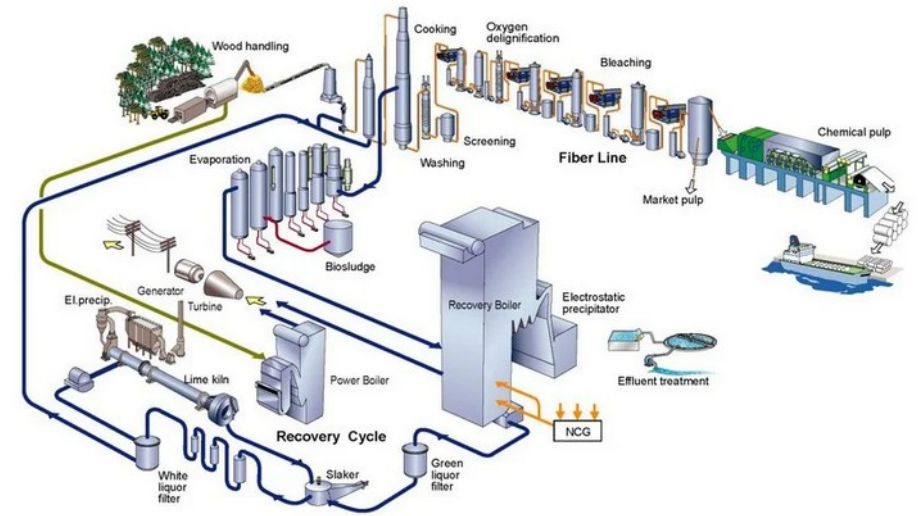
Discrete Manufacturing and Continuous Process Industry

DISCRETE OR MACHINE AUTOMATION



- Conveyors
- Packaging machines
- Robot cells
- Logistics systems
- Individual products or items

CONTINUOUS PROCESS INDUSTRY



- Power plants
- Pulp and paper mills
- Chemical plants
- Water and energy processes
- Changing flows and variables

DISCUSS | Which environment usually requires continuous operator supervision?

Continuous Operation Changes the Requirements

PLC-ORIENTED MACHINE



- Runs automatically for long periods
- Local HMI supports setup and disturbances

DCS-CONTROLLED PLANT



- Continuous monitoring
- Setpoints and modes
- Alarms and trends
- Several process areas

DISCUSS | What other kind of differences do you think there is?

PLC and DCS Compared

PLC and DCS technologies overlap. These are typical tendencies, not absolute technical limits.

Aspect	Typical PLC solution	Typical DCS solution
Application focus	Machine or compact automation	Large and complex process automation
Typical example	Package handling or a production machine	Power plant, pulp mill, refinery, dairy, oil rig, ship
System size	Often a smaller I/O count	Often a larger I/O count
Operation	May run autonomously or with limited HMI use	Continuous operator interaction is commonly central
Architecture	Often centred on one or a few controllers	Functions distributed across several stations and nodes
Expansion	Can expand, depending on platform and design	Designed to scale through stations, nodes and subsystems

KEY IDEA | Compare application context, operator role, lifecycle and integration — not only I/O count.

PLC and DCS Perform the Same Core Control Tasks

SHARED CONTROL CYCLE



TYPICAL PLC SYSTEM

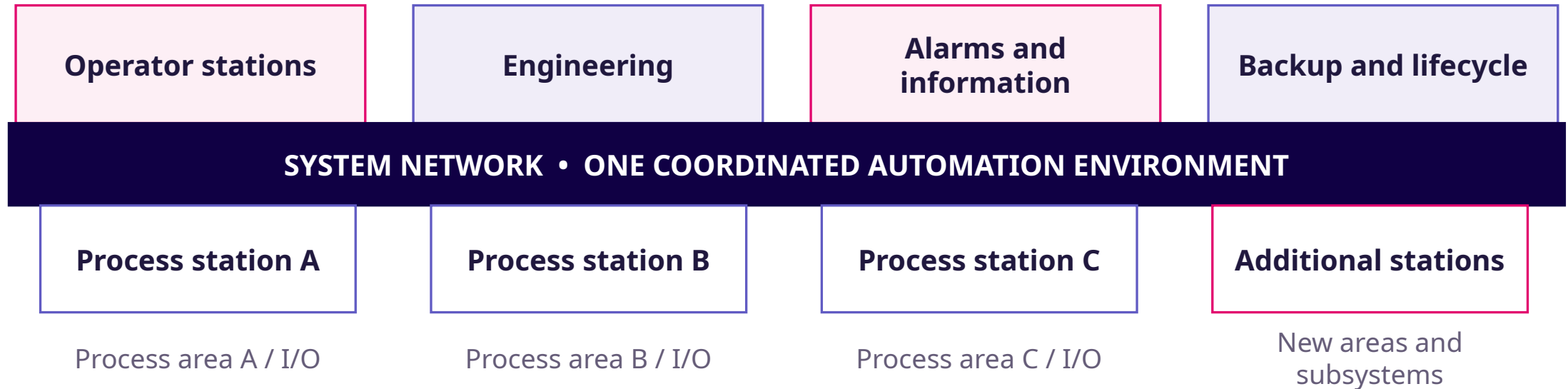
Control is often centred on one or a few controllers.

TYPICAL DCS SYSTEM

The same tasks operate within a distributed, plant-wide system.

Same automation principles — different system organisation, scale and lifecycle support.

DCS Distributes Functions to Scale as One System



Functions are distributed between specialised stations; capacity grows by adding stations, nodes and process areas.

KEY IDEA | Distribution supports expansion while the environment remains one coordinated system.

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Scalability Means Managing the Whole System

ONE COORDINATED SYSTEM

Operator workplaces • Engineering • System management

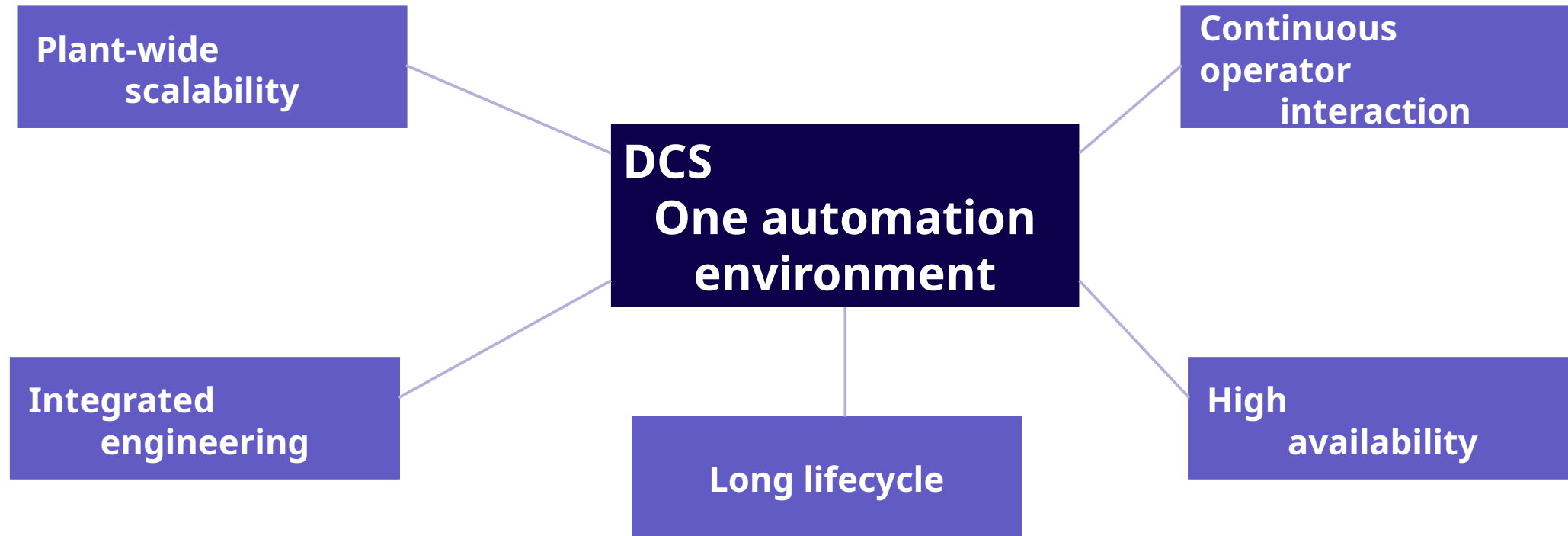
Shared alarms • Trends • Process information

Process area A Process area B Process area C

Distributed field I/O and industrial communication

A DCS scales by combining specialised components into one coordinated automation system.

Why Do Distributed Control Systems Exist?



A unified environment for large, continuously operated processes.

Valmet DNA as the Course Platform



- Concrete example of distributed process automation
- Finland has strong roots in DCS business ->
- Focus: concepts, system roles and application engineering
- Principles apply to other manufacturers' systems:

Siemens PCS 7



ABB Ability System
800xA



Honeywell Experion PKS
/ Alcont

Honeywell

Emerson DeltaV



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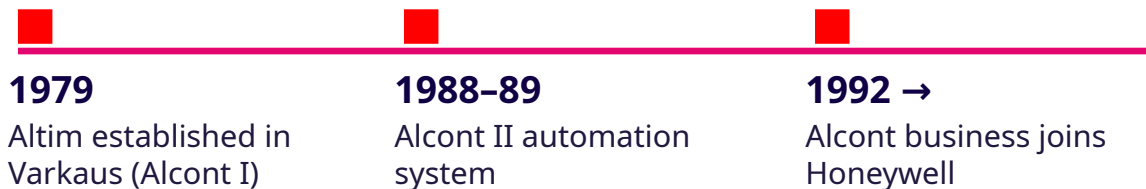
Finland: A Pioneer in Distributed Control Systems

Finland did not only adopt DCS technology — Finnish industry also helped to develop it.

VALMET / METSO • DNA



ALTIM CONTROL • HONEYWELL ALCONT



Strong process industry

Long industrial plant lifecycles

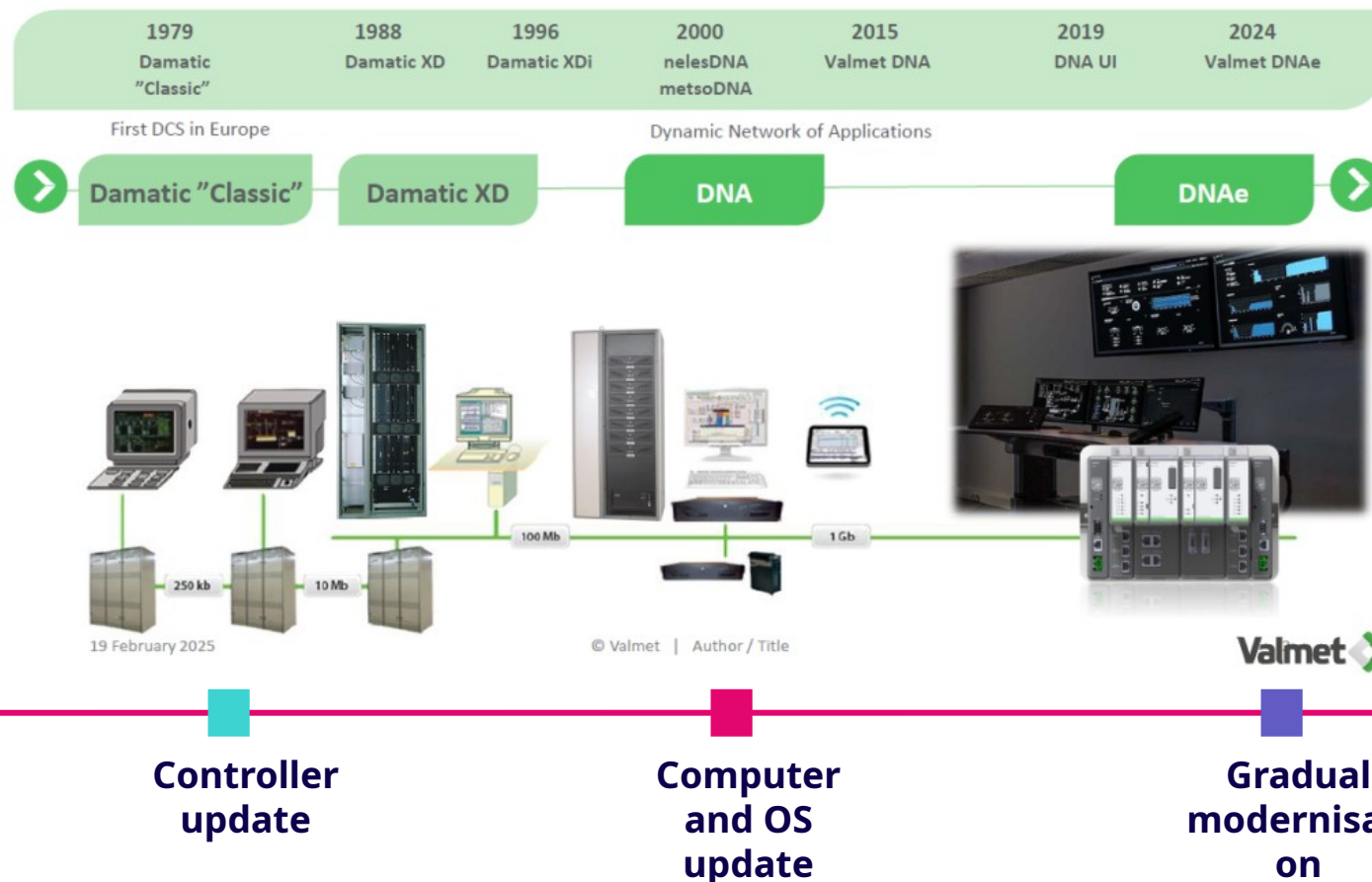
Continuous development of Finnish automation technology

DISCUSS | Why did Finland become an important country in the development of DCS technology?

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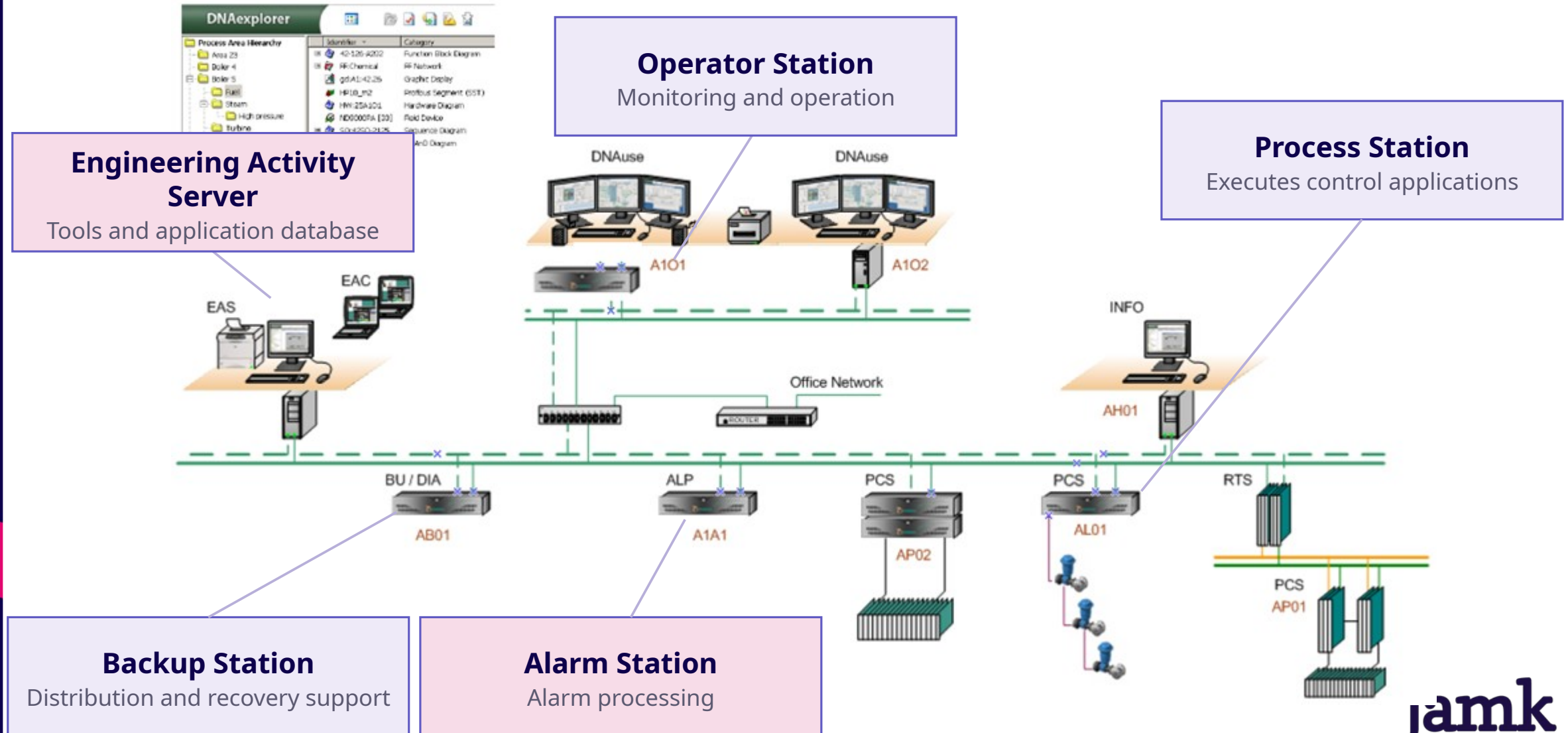
Automation Lifecycle

The industrial process continues for decades



DISCUSS | Why is replacing the complete automation system at once often difficult in a continuously operating plant?

DCS Functions Are Divided Between Specialised Components



Another Perspective on Distribution and Scalability

Compact solution

A controller, I/O and several server functions combined in a compact industrial computer.

Laboratory or medium solution

Dedicated process station and physical I/O combined with virtualised server functions.

Large industrial solution

Multiple subsystems, process stations, operator environments and process areas connected into one automation system.

Tiny embedded application

- 10 - 250 IO's
- typically installed to field or embedded to machines

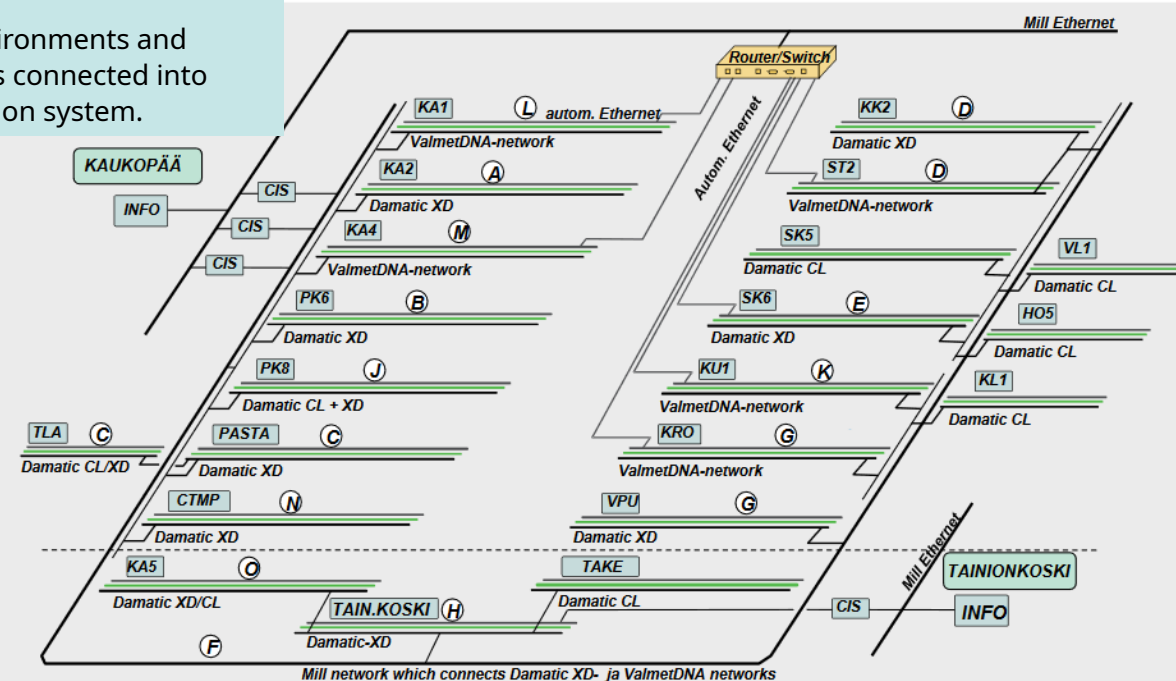


Operator Interface

- Valmet DNA Operate
- Alarm processor

ACN SR1 and IO

- ACN SR1
- ACN IO
- Backup functionality



Functions are distributed, but the complete environment remains coordinated.

DISCUSS | What are the benefits of DCS system scalability for automation projects and maintenance?

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Connecting DCS Concepts to the PLC World

FAMILIAR PLC CONCEPT

PLC CPU

TIA Portal engineering project

HMI or SCADA

Device diagnostics

Distributed I/O and fieldbus



RELATED DCS CONCEPT

Process Station

Centralised engineering environment

Operator Station

Integrated system diagnostics

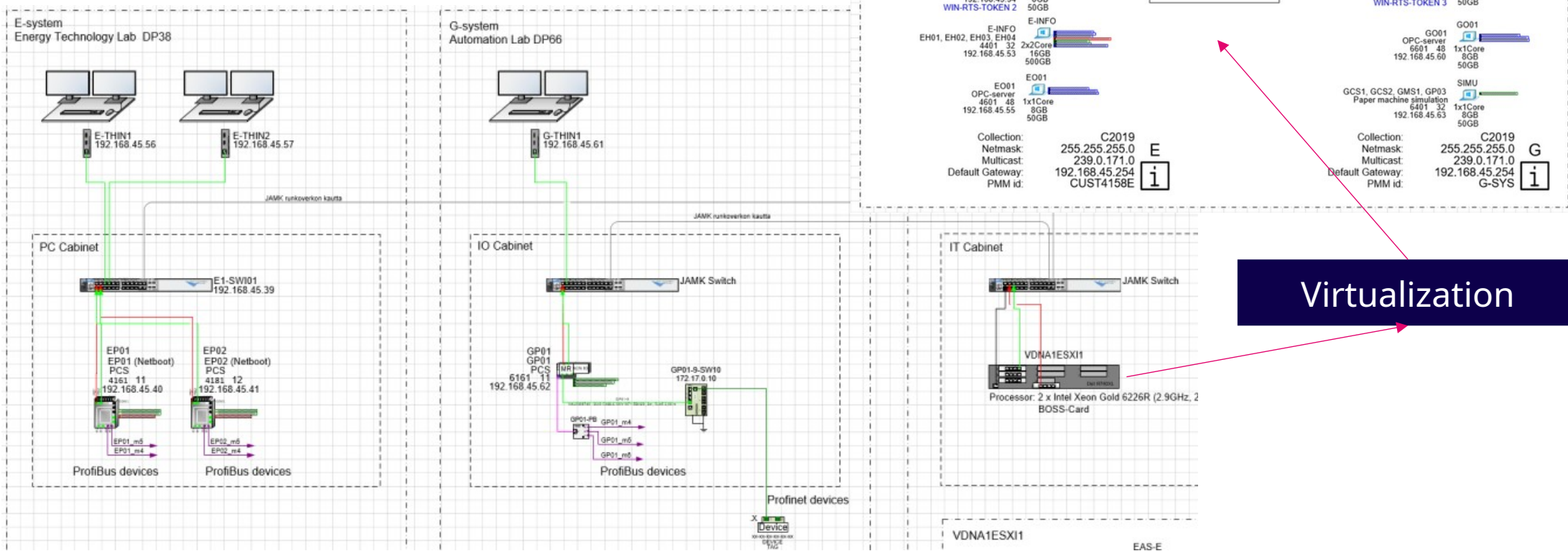
Distributed process I/O and fieldbus

These are conceptual comparisons, not exact one-to-one equivalents.

DISCUSS | Choose one pair and explain one similarity and one difference.

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Jamk Valmet DNA Environment



Virtualization

Specialised stations are connected into one automation environment.

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Would You Choose PLC or DCS?

Select PLC, DCS or “depends” — justify with operation, integration, scale and lifecycle.

1	Standalone packaging machine	PLC / DCS / DEPENDS
2	Small pumping station within plant	PLC / DCS / DEPENDS
3	Waste Water Treatment plant	PLC / DCS / DEPENDS
4	Continuously operated power plant	PLC / DCS / DEPENDS

Use at least two criteria in every justification.

Could an Entire Process Plant Be Controlled with PLCs?

Technically, yes. The challenge is operating, integrating and maintaining the complete plant as one system.

SEPARATE PLC AREAS

Process area A

PLC • engineering project • local HMI

Process area B

PLC • engineering project • local HMI

Process area C

PLC • engineering project • local HMI

Possible — but independently engineered and operated.

Several process areas

Thousands of field signals

Continuous 24/7 operation

Multiple operators



Central alarm management

Shared process information

Integrated engineering

Decades-long lifecycle

COORDINATED DCS

AREAS A • B • C

- **Distributed control**
- Shared operator interface
- Central alarm management
- Integrated engineering
- Common process information

THINK-PAIR-SHARE | 2 MINUTES

Which problems might occur if every process area were implemented as an independent PLC project?

A DCS turns distributed process areas into one coordinated plant-wide automation system.

Five Ideas to Remember

- 1 PLC and DCS perform the same fundamental control task.
- 2 DCS is designed for integrated, plant-wide process automation.
- 3 Continuous operator interaction is a central DCS requirement.
- 4 Distributed specialised components improve scalability and management.
- 5 Long lifecycle and gradual modernisation influence DCS architecture.

DISCUSS | Explain in your own words why a DCS suits a continuously operating process plant.

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